PointDev: Open Structured Context Capture

Project detail attachment for NLnet NGI Zero Commons Fund application

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118	1,188	86	v0.1.0
COMMITS	UNIT TESTS	SOURCE FILES	TAGGED RELEASE

The Problem

MCP, function calling, and tool use standardised how AI agents act on the world. No equivalent exists for the other direction: humans telling agents what they want with structural precision.

A developer types "the hero font is too small." A client sends a Loom recording. A QA engineer files a ticket: "the text on the homepage? The big one?" Each message drops the DOM path, computed styles, component name, and spatial intent. Proprietary AI tools are building their own closed context capture. Without an open format now, structured human-to-AI communication locks into walled gardens.

Real Capture Output

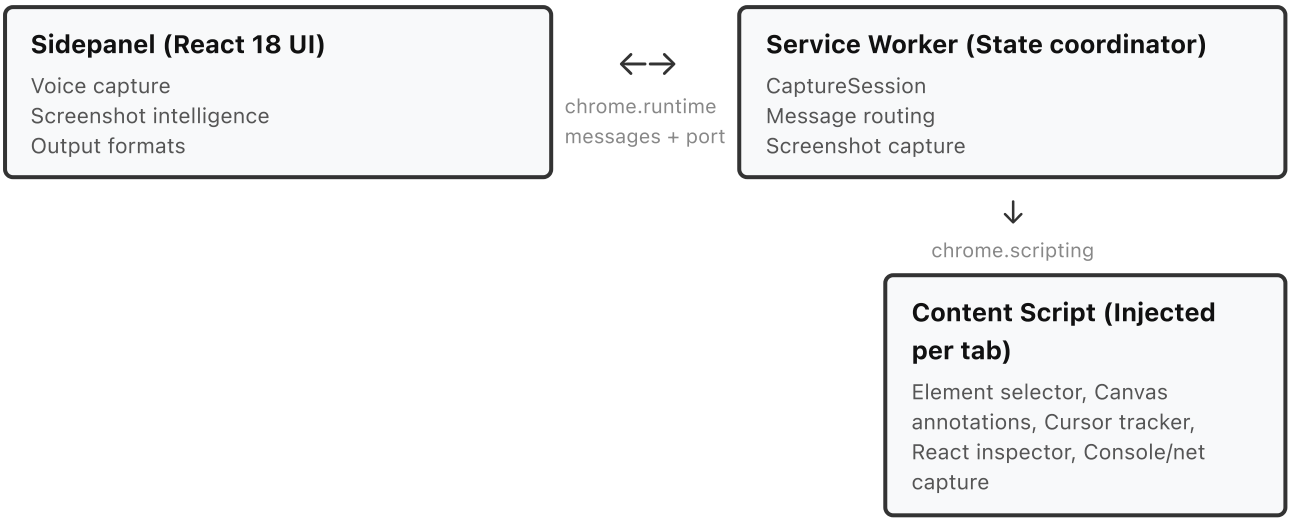
Output from a capture session on a live site. Voice narration, spatial annotations, and cursor dwell data fused with DOM context:

```
[00:06] "main hero is far too large"
[00:23] "the scroll CTA is overlapping with the subtitle"
1. [00:40] Circle around .lg\:min-h-\[100svh\] at (42, 1019)
   - Dwelled 25.5s over h1.font-display.text-hero (during: "main hero")
   - Dwelled 7.3s over div.absolute.bottom-10 (during: "scroll CTA")

Element: h1.font-display.text-hero
  React: <HeroHeading>
  Selector: main > section > h1
  Computed: font-size: 72px; line-height: 1.1; color: rgb(15, 23, 42)

Console errors: 1
  [00:02] TypeError: Cannot read properties of undefined (reading 'map')
    at ProductGrid.tsx:42
```

An AI agent reading this knows which elements the user means, what they want changed, and where they were looking while saying it.

Architecture

Chrome MV3 extension with four cooperating contexts:



Message flow: Sidepanel ↔ Service Worker ↔ Content Script

What the PoC Implements

- **Bridge server + MCP tool handlers:** WebSocket bridge delivers captures to AI tools, with handlers for `get_session`, `get_voice_transcript`, `get_annotations`, `get_screenshot`
- **Three output formats:** plain text, JSON, Markdown with pluggable compiler architecture
- **Console and network capture:** monkey-patches `console.error`, `console.warn`, `fetch`, `XMLHttpRequest` in the MAIN world via script injection (avoids chrome.debugger warning bar)
- **Screenshot intelligence:** interest-scoring using frame differencing (30%), cursor dwell (25%), voice activity (35%) to auto-capture at moments of user attention
- **Annotation tools:** circle, arrow, freehand, rectangle with scroll anchoring
- **On-device Whisper transcription:** Transformers.js + Xenova/whisper-tiny.en, functional but needs production hardening
- **Cursor dwell tracking:** temporal correlation to voice segments

What This Grant Funds

Six milestones covering the remaining R&D to take a working PoC to production-quality open infrastructure:

#	Milestone	Days	Cost
M1	**Local Whisper transcription, production hardening.** AudioWorklet migration, model download/caching UX, memory profiling, streaming audio exploration.	~20	€6,000
M2	**Open structured output format specification.** Version the format, write spec document, align with W3C Web Annotation / MCP schemas, publish.	~12	€3,600
M3	**Accessibility context capture.** ARIA roles, a11y tree, contrast ratios, WCAG violation detection. Adapt pi-annotate patterns (MIT).	~14	€4,200

M4	**Cross-framework detection (Vue + Svelte).** vue-inspector.ts, svelte-inspector.ts following react-inspector.ts pattern, production build fallbacks.	~12	€3,600
M5	**Firefox cross-browser port.** WebExtensions compatibility layer, sidebar_action alternative, CSP differences, testing.	~10	€3,000
M6	**Prompt compiler npm package + docs + community.** Extract compiler as standalone lib, developer docs, test coverage, MCP ecosystem outreach, demo video.	~12	€3,600

Total: ~80 days (€24,000) + infrastructure/CI (€500) + contingency (€500) = **€25,000**

Rate: €300/day (solo developer, cost-recovery; commercial rate: €1,200/day). Self-funded to date.

Expected Outcomes

- A privacy-first browser extension with local voice transcription, working across Chrome and Firefox, detecting React/Vue/Svelte components, capturing accessibility context
- A versioned, documented open format specification for structured context capture, the input-side counterpart to MCP, that any AI tool, accessibility auditor, or UX researcher can implement
- A standalone prompt compiler library (npm) that merges technical and human context with temporal correlation
- An MCP-compatible bridge server published as a formal MCP tool for Claude Code, Aider, and others
- A format designed for sharing between stakeholders. The collaborative sharing infrastructure (relay, viewer, team features) is the trajectory for a follow-up grant

If other tools adopt the format, PointDev succeeded. The extension is one implementation.

PointDev is MIT licensed. Repository: github.com/BraedenBDev/pointdev
AI-assisted development disclosed in repository: Co-Authored-By tags on all AI commits, per-session development log at docs/genai-disclosure/development-log.md